EnergyBench: a live, gate-aligned benchmark of day-ahead forecasts for European power markets

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energybench.foretec.co · github.com/ujohn33/foretec-energy-bench

Abstract

Energy forecasting models are usually compared on backtests and fixed public datasets. Backtests can use data that did not yet exist when the forecast would have been made, and a fixed test set, reused across studies and possibly contained in the pretraining data of foundation models, invites overfitting to the benchmark. EnergyBench evaluates every entry live instead. Each day at 11:30 CET/CEST, thirty minutes before the day-ahead (SDAC) gate closes, every model forecasts the next day's price, wind and solar generation and load on the 15-minute grid for six bidding zones (BE, NL, FR, DE-LU, DK1, DK2). Unlike existing live benchmarks, EnergyBench runs the models itself, on one information set that it freezes at the cut-off and publishes, including three numerical weather prediction models, so that foundation models can be compared with and without the same weather covariates under the actual market deadline. Errors are pooled per series against a seasonal-naive baseline, and models are ranked by a Bradley–Terry rating on the Elo scale with a day-bootstrap interval. The field spans baselines, linear and gradient-boosted models on the frozen weather, eighteen time-series foundation models from eleven families, ensembles, and the grid operators' own forecasts. Over the first thirty live days, the covariates reduced the error of each of the ten foundation models run with and without them by 38–52% on matched races (64–72% on wind, 55–62% on solar). Used zero-shot, these models match or outperform a LightGBM model trained on the same covariates. Model size, from 3M to 331M parameters, has a much smaller effect, and median ensembles of the covariate models rank highest. The grid operators' pre-gate forecasts remain ahead on wind and solar, where the best single foundation models have 20–28% more error; on load the two are level.

1 Introduction

Day-ahead forecasts of price, renewable generation and load are central inputs to bidding decisions on European power markets, yet reported accuracies of different models are rarely comparable [5, 21]. Three problems commonly affect backtests [5]. First, the information set drifts: a forecast nominally made at 11:30 is built from weather runs that were published hours later, from actuals that were still provisional, or from a TSO forecast that only appears after the gate [18]. Second, the target may differ from the quantity the market trades, for example hourly values where the market now trades quarter-hours, generation definitions that differ between data providers, or actuals that were later revised. Third, the evaluation window, the baseline and the metric can be chosen after the results are known. Fixed public datasets add a related problem: when the same test set is reused across studies, models and settings are

selected on it repeatedly, and for foundation models the test period may also be part of the pretraining data [4, 15].

A fourth problem concerns the evaluation of time-series foundation models. In operational energy forecasting, exogenous inputs are the standard, not an option: wind and solar forecasts are built on numerical weather predictions, load forecasts on temperature and calendar effects, and day-ahead price forecasts on forecasts of load, renewable infeed and fuel cost [19–21]. The grid operators' own forecasts, the industry reference, are NWP-driven, and the energy forecasting competitions routinely supplied weather forecasts [16]. The benchmarks on which foundation models are compared are, by contrast, mostly univariate; fev-bench [13] and, for energy, FETS [14] are recent exceptions. A model that forecasts tomorrow's solar output from the last four weeks of solar output alone is evaluated on a task that does not arise in practice, and a ranking obtained in this way is of limited use for choosing a model to deploy.

A live benchmark that provides every entry with the same inputs avoids the first two problems and the risk of contamination. If each model runs at the actual cut-off on the data available at that time, weather forecasts included, and its forecast is locked before the actual gate, look-ahead is excluded. Because the outcomes do not yet exist when the forecasts are made, no model can have been trained or tuned on them, and the record grows daily and is public. The drawback is that a live record is short and its standings vary from day to day. The design described below accounts for this variability. Errors are pooled before they are normalised, rankings are reported with intervals, and only live forecasts are reported. The third problem, the choice of analysis after the results are known, is reduced but not removed by fixed public rules (Section 5).

EnergyBench began operating on 2 October 2026 with three zones (BE, NL, FR) and three targets; total load was added on 6 October and three zones (DE-LU, DK1, DK2) on 15 October. The code and the configuration file that define the benchmark are versioned in a public repository. This paper describes that configuration as of 31 October 2026 and the results to date.

2 Related work

Static benchmarks. Forecasting competitions have shaped the field since the M-competitions [1]; the Monash archive [2], GIFT-Eval [3] and fev-bench [13] standardise the evaluation of general-purpose and foundation time-series models on fixed datasets; 46 of fev-bench's 100 tasks include covariates. In energy, the Global Energy Forecasting Competitions [16] posed realistic tasks as one-off events, and Lago et al. [21] published an open benchmark for day-ahead price forecasting with fixed splits and day-ahead exogenous forecasts for five markets. FETS [14] compares foundation models, with and without covariates, against task-specific gradient-boosted and random-forest models on 54 public energy datasets in nine categories, with archived ECMWF IFS forecasts as weather covariates over the forecast horizon, and finds covariate-informed zero-shot foundation models best in aggregate. Fixed datasets have two weaknesses for the present purpose. They describe a past regime, and energy systems change quickly; and their test periods may have been seen, directly or through pretraining corpora, by the models they evaluate. Meyer et al. [4] document both direct overlap, where one model's training data serve as another's test set, and temporal overlap between correlated training and test series in the evaluation of foundation models. Retrospective energy studies also often leave the information set implicit: several price-forecasting studies use ENTSO-E's day-ahead wind and solar forecasts as inputs although those are published only after 18:00 on $D-1$, six hours after the auction closes [18].

Covariates. The value of weather inputs in energy forecasting is long established. Wind and solar power forecasting has been NWP-based since its beginnings, and the wind and solar tracks of GEFCom2014 gave every team ECMWF forecasts [16]; load forecasting rests on temperature, and the reviews of price forecasting single out forecasts of load and renewable generation as the most informative exogenous drivers [19–21]. Foundation models have only recently gained the means to use such inputs: Chronos-2, TimesFM, TiRex-2, Timestral, TimeH2O and the t0 models accept known-future covariates,

by different mechanisms and with different success. Whether and how much they benefit cannot be measured on a univariate benchmark.

Live benchmarks. Three recent benchmarks score forecasts made before the outcome is known; the two that cover energy are compared with EnergyBench in Table 1. TS-Arena [17] pre-registers forecasts of time-series foundation models for German, US and Finnish price, load and generation series and ranks them by an online Elo rating. It provides no covariates, so the models' use of weather forecasts, which in our results reduces their error by 38–52% (Section 4), is not assessed; and each round starts from a series' last published value rather than at a market gate, so the forecasts do not correspond to bids that could have been placed. Energy-Arena [18] is gate-aligned: forecasts for the next day are submitted before the 12:00 day-ahead gate. It compares submissions rather than models under common conditions, since participants run their own models on inputs of their choice, which can be neither verified nor reproduced. Impermanent [15], from the TimeCopilot project, runs the models itself and scores them as the observations arrive, motivated by the contamination of static test sets and by model selection on them; its series are univariate counts of GitHub activity rather than energy series. In industry, the Predico platform [25] scores live renewable forecasts submitted to Elia and 50Hertz against the real gate, for enrolled forecasters and without publishing their inputs.

Positioning. EnergyBench takes Energy-Arena's operational framing (the day-ahead gate defines the task) and TS-Arena's pre-registration logic (only a forecast made before the outcome is known is free of leakage), and, like Impermanent, runs every model itself, on an information set that includes weather forecasts. Each day it fetches one information set at the gate-aligned cut-off, freezes and publishes it, and gives it to every entry: the information set is enforced rather than declared, and any forecast can be reproduced from its archived inputs. Foundation models from seven families also run with the same frozen covariates, so the effect of the weather inputs is measured for each model; all four targets are forecast in six zones; and the grid operators' own forecasts, before and after the gate, serve as references. In return, entries must be submitted as code rather than as forecasts, and proprietary data cannot be used. Table 1 summarises the differences.

3 The benchmark

This section defines the benchmark as it runs: the forecasting task, the daily timeline and what a model may know at the cut-off, where the actuals come from, the covariates frozen for every entry, the field of entries, and how forecasts are scored, ranked and valued.

3.1 Task

For each zone $z \in \{\text{BE}, \text{NL}, \text{FR}, \text{DE-LU}, \text{DK1}, \text{DK2}\}$ and target $y \in \{\text{price}, \text{wind}, \text{solar}, \text{load}\}$ (24 series), a forecast made on issue day $D-1$ covers every quarter-hour of delivery day D in local time: 96 values, or 92 and 100 on clock-change days. Price is the SDAC day-ahead price (EUR/MWh); wind is onshore plus offshore generation, solar is solar generation and load is total load (MW). Models return a point forecast and optionally the nine deciles $q_{0.1}, \ldots, q_{0.9}$.

3.2 Timeline and information set

The benchmark day is defined by three clock times in CET/CEST (Figure 4). At the *cut-off*, 11:30 on $D-1$, a snapshot of everything public is taken and every model runs on it. The *gate* is 12:00, the SDAC gate closure. A live run that finishes after the gate is marked late and never scored, and a day's forecasts cannot be changed once written, so the record consists of the forecasts as they were at the gate. Prices are scored on $D-1$ after the auction for D has cleared, at the scoring run of 14:45, or at that of 20:45 if the prices were not yet published. Wind, solar and load are scored at 07:00 on $D+1$ if every quarter-hour of D has been published, and otherwise at the first later scoring run at which 95% has.

Table 1: Live energy forecasting benchmarks compared, as described in their papers and on their websites (October 2026).

	TS-Arena [17]	Energy-Arena [18]	EnergyBench (this paper)
Task timing	from each series' last published value; no market gate	submission before the 12:00 day-ahead gate, whole next day	11:30 cut-off, locked before the 12:00 gate, whole next day
Zones and targets	14 challenges (SMARD, Gridstatus, Fingrid)	wind, solar, load: AT, DE-LU; price: 15 zones	price, wind, solar, load in each of BE, NL, FR, DE-LU, DK1, DK2
Who runs models	participants (containers)	participants (API submissions)	the benchmark, from submitted code
Information set	series history only	anything before the gate; self-declared, unverified	frozen and published: history, three NWP models, load forecast, fuel
Covariates for foundation models	none	if a participant adds them	eleven foundation models also with the same covariates
Foundation models	the focus of the benchmark	none named; depends on submissions	eleven families, and ensembles
Transparency	platform code public	method description and forecasts optional	code, inputs and forecasts public
Scores	MASE	RMSE, WIS, CRPS	MAE, RMSE, bias, pinball; relative to seasonal naive, pooled per series
Ranking	online Elo, replay bootstrap	per metric over rolling windows	Bradley–Terry Elo, day bootstrap
References	statistical baselines	optional benchmark models	baselines; the TSOs' forecasts before and after the gate

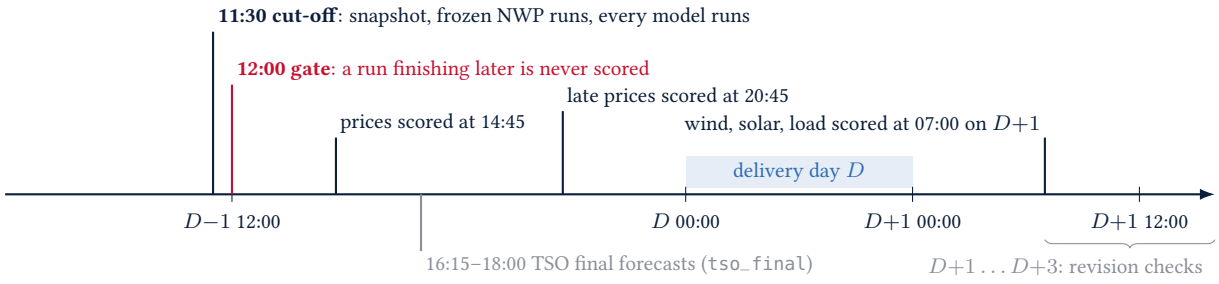


Figure 1: One benchmark day (not to scale). Everything a model sees is fixed at the 11:30 cut-off; the forecast must be locked before the 12:00 gate. The TSOs' final day-ahead wind and solar forecasts appear only after the gate and enter as references.

What a model may know at the cut-off is set per target. Prices: everything up to the end of $D-1$, which the auction on $D-2$ published. Wind, solar and load: actuals up to the cut-off minus one hour of publication lag. Every model receives the last 28 days of 15-minute history (the context). A model may train on longer history, provided that each training row uses only data from before that row's own cut-off. Two qualifications apply: the training actuals are the current values, which may have been revised since they were first published, and the weather in the two-year training frames of two entries is matched to the live lead time rather than to the live run (Section 3.4). Training never takes place inside the live window. Models that retrain do so in the nightly catch-up at 03:30. The TSOs' day-ahead renewable forecasts are not an input: Regulation (EU) 543/2013 [26] has them published by 18:00 on $D-1$, after the gate.

Table 2: Primary source of the actuals per zone and target. Gaps are filled from the TSO's own data (Elia, RTE éCO2mix), then from Energy-Charts; the Dutch NED series have no fallback.

	Price	Wind	Solar	Load
ENTSO-E document	A44	A75 (B18 + B19)	A75 (B16)	A65
BE	ENTSO-E	Elia (quarter-hourly)	Elia (quarter-hourly)	ENTSO-E
NL	ENTSO-E	NED (national)	NED (national)	NED (gross)
FR	ENTSO-E	ENTSO-E	ENTSO-E	ENTSO-E
DE-LU	ENTSO-E	ENTSO-E	ENTSO-E	ENTSO-E
DK1, DK2	ENTSO-E	Energinet	ENTSO-E	ENTSO-E

3.3 Actuals and provenance

The primary source is the ENTSO-E Transparency Platform [27], parsed from its XML: day-ahead prices (document A44, the 15-minute SDAC prices), actual generation per production type (A75; wind as onshore B19 plus offshore B18, solar B16) and actual total load (A65). ENTSO-E does not deliver every quarter-hour: alone it lacked 12% of Belgian and 2.6% of French wind quarter-hours in September 2026. Gaps are filled from the TSOs' own open data (Elia for Belgium, RTE éCO2mix for France), then from Energy-Charts. Where a TSO publishes finer data than it sends ENTSO-E, or a quantity closer to the national total, the TSO is the primary source (Table 2). The source of every value is recorded.[1]

In the Netherlands, ENTSO-E's series cover only what TenneT meters itself: about 1% of the solar fleet (a capacity factor of 0.1% over 20 September–7 October 2026, against 10.7% in the national estimate), mostly offshore wind, and load net of embedded solar (about 6 GW against 17 GW gross at a sunny midday). The Dutch wind, solar and load targets therefore come from NED, the Nationaal Energie Dashboard of TenneT and the distribution operators, which publishes national estimates: wind as onshore plus metered offshore, load as gross consumption as Elia and RTE report it. These series have no fallback, since filling a gap from ENTSO-E would combine two definitions within one delivery day; quarter-hours that NED has not published are left unscored.[2]

Actuals are *frozen* the first time they are complete and scored from the frozen copy. Until $D+3$, wind, solar and load are re-fetched at every scoring run; a change above 0.5% of the day's energy re-freezes the actuals and re-scores the day, and every check is logged. Prices are final at the auction. Between the gate and scoring, the site shows the quarter-hours metered so far as provisional actuals, which are never scored.

3.4 Covariates frozen at the cut-off

Models that declare covariate inputs receive, per issue day, a table frozen at the cut-off and a log of what went into it.

Weather. Three independent global and regional NWP models, ICON-EU (DWD), GFS (NOAA) and IFS HRES (ECMWF), are fetched run by run from the Open-Meteo single-runs archive [28]. A run is admitted only if its initialisation is at most the cut-off minus a lag, the provider's measured publication delay plus a margin (3.5 h for ICON-EU, 7 h for GFS and IFS); at 11:30 this admits the 06 UTC ICON-EU run and the 00 UTC GFS and IFS runs, in summer and in winter time (Figure 2). Since 19 October a run must also be complete: if any variable is less than 99% filled over lead times 12–48 h, which contain the

[1] The sources for DE-LU, DK1 and DK2 follow those of Electricity Maps [7], except for Danish wind: ENTSO-E's DK1 offshore series covers only the parks registered there and has missed 0.3–0.5 GW of offshore capacity since July 2026, so Danish wind has come from Energinet since delivery day 18 October. The first two issue days, 2 and 3 October, were scored on Energy-Charts data alone.

[2] Dutch forecasts made against the earlier ENTSO-E definition (issue days up to 10 October for wind and solar, 11 October for load) are not scored. NED's offshore wind was a park-by-park model through delivery day 12 October, 7–27% above the metered output used since; those days keep their scores.

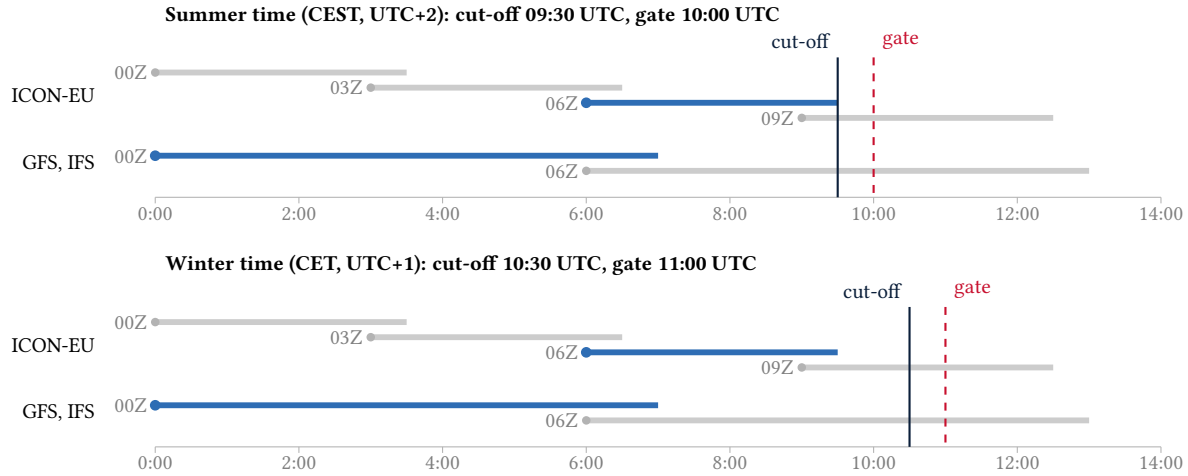


Figure 2: NWP runs admitted at the cut-off, in UTC. Each bar runs from a run's initialisation (dot) to the time it is admitted, initialisation plus the lag (3.5 h for ICON-EU, 7 h for GFS and IFS); blue: the run the covariates use. The cut-off stays at 11:30 local time, so it moves by an hour in UTC with the clock change, but the same runs are admitted in both regimes: ICON-EU 06 UTC (exactly at the summer cut-off) and GFS and IFS 00 UTC.

Table 3: Weather sampling points per zone and bucket (capacity-weighted k-means centroids; load weighted by population) and the public registers they are built from.

Zone	Onshore wind	Offshore wind	Solar	Load	Registers
BE	8	2	8	6	OpenStreetMap turbines scaled to Elia's capacity; RIVM offshore turbines; Fluvius and ORES municipal PV scaled to Elia; Statbel population
NL	8	4	8	6	RIVM turbines; CBS municipal PV and population
FR	8	4	8	6	ODRÉ national register of installations (commune level); INSEE population
DE-LU	8	4	8	6	Marktstammdatenregister; GeoNames population
DK1	8	4	8	6	Energinet capacity per municipality; Danish Energy Agency offshore farms; GeoNames population
DK2	8	3	8	6	as DK1
Total	48	21	48	36	153 points

delivery day, the fetch is retried up to five times and the builder then falls back to an earlier cycle, at most two (Section 5). Weather is sampled at 153 capacity-weighted k-means centroids built from public registers (Table 3). The table carries, per variable and bucket, each NWP model's capacity-weighted zone mean, and across the three models the ensemble mean, spread and pairwise differences, so that disagreement can serve as a feature. Models trained on the 28-day context train on tables built by the same rule day by day, from the runs that were available at each day's own cut-off. The two-year training frames of `lgbm_prod` and `lin_cov_2y` predate the single-runs archive for ICON-EU and GFS; there the archive's previous-runs product is matched to the live lead time to the nearest day, which gives forecasts of the same lead but not always of the same run (Section 5). Figure 3 shows the sampling points.

Load forecast. ENTSO-E's day-ahead total load forecast (A65), which TSOs must publish two hours before the gate. It is an input of the price models. By design the load models do not receive it, because in most zones it is itself the pre-gate reference they are compared with (Section 4.6); until issue day 15 October they did. Section 5 records a parsing fault in its Danish series.

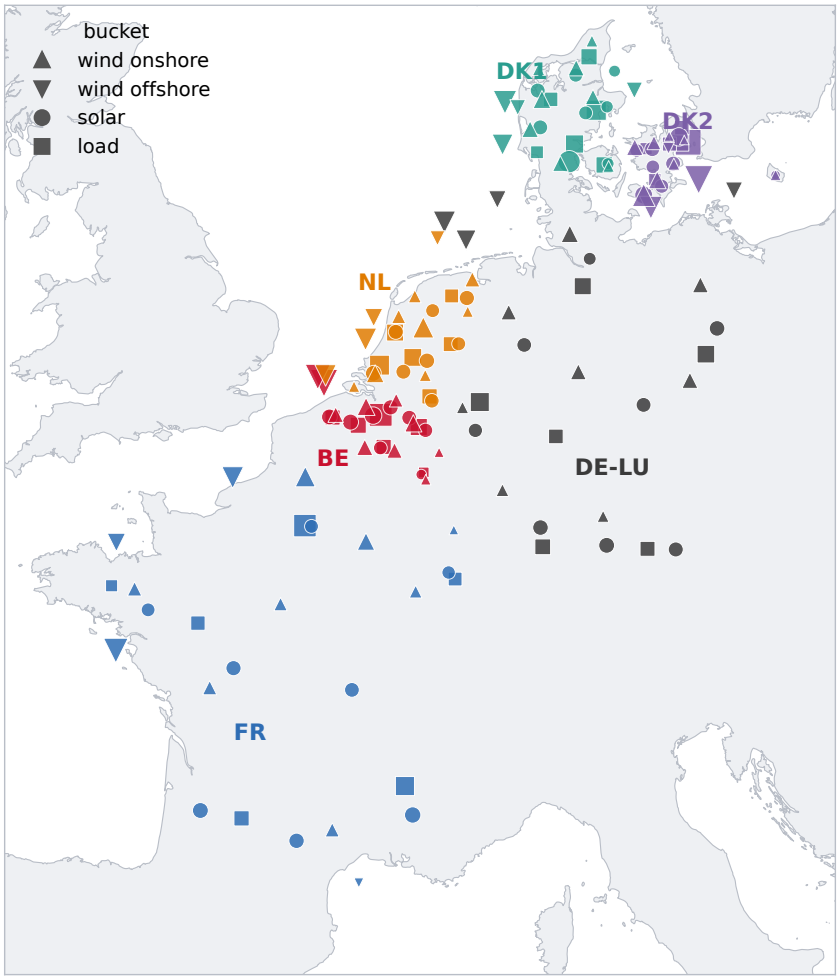


Figure 3: The 153 weather sampling points: capacity-weighted k-means centroids per zone and bucket, from public registers (load: population). Marker size is the point's share of its zone and bucket.

Fuel cost. Daily closes of TTF gas (ICE Dutch front month) and of an exchange-traded EU carbon allowance certificate, as published by Yahoo Finance, the last close dated two days before delivery or earlier, and the derived short-run marginal cost of a combined-cycle gas plant in EUR/MWh,

$$\mathrm{SRMC} = \frac{P_{\mathrm{TTF}}}{\eta} + e\,P_{\mathrm{EUA}}, \qquad \eta = 0.52, \quad e = 0.37\,\mathrm{t\,CO_2/MWh}, \tag{1}$$

the fuel and carbon cost in the clean spark spread, with η the plant's efficiency and e its emission intensity per MWh of electricity.

3.5 Entries

A model is a folder with a declarative `model.yaml`: a named sktime [29] estimator, a Python module, or a subprocess script with its own environment, the route all foundation models take (each model loads its weights once per run, for all series). A model whose environment is missing is skipped and logged, not failed. The most recent live run, on 4 November 2026, took 17 minutes for the whole field and all 24 series on a four-core, 8 GB server, within the 30-minute window; two models run in parallel. The field on 31 October (Table 4) has 43 active entries.

No TSO in France or Germany publishes a day-ahead wind or solar forecast before the gate (RTE at 16:15, the German TSOs at 18:00 on $D-1$), and Energinet's is not verified as published before it, so `tso_cutoff` has wind and solar only for BE and NL. Classical univariate models (ARIMA, ETS, Theta,

Table 4: The field, 31 October 2026 (43 active entries). Parameter counts are those of the checkpoints the benchmark loads. Covariates: *native*, the model's own known-future covariate input; *ridge*, an in-context ridge regression on the covariates whose residual the model forecasts; *full*, the whole covariate table.

Entry	Params	Covariates	Developer, set-up
Baselines			
`naive_weekly`			same quarter-hour one week earlier; the reference of all relative errors
`naive_daily`			same quarter-hour one day earlier
`climatology`			same local time of day within ± 14 days, 2023 to August 2026
`lin_lags`, `lgbm_lags`			ridge and LightGBM on the series' own lags; two years, retrained monthly
Models trained on the covariates[a]			
`lin_cov`, `lgbm_cov`		full	ridge and LightGBM on the 28-day context
`lin_cov_2y`		full	ridge on two years
`lgbm_prod`		full	LightGBM on two years, 120-day recency half-life, solar on its capacity factor[b]
Foundation models, zero-shot[c]			
Chronos-2 [30]	120M	native	Amazon
Chronos-Bolt base [30]	205M		Amazon
TimesFM 2.5 [31]	231M	ridge	Google
TimesFM 3.0 [31]	331M	native	Google
Moirai 2 small [32]	11M	[d]	Salesforce
TiRex [33]	35M		NX-AI
TiRex-2 [33]	95M	native	NX-AI; about 38M active in univariate mode
Toto	151M	[d]	Datadog; retired 27 October, superseded by Toto 2
Toto 2	313M, 1B		Datadog; live from 25 October
Sundial	128M		Tsinghua University
TTM r2	3.4M		IBM
Tabby, Tabby-Pretrain	147M, 146M	ridge	Huawei Noah's Ark Lab; load forecast without covariates
Timestral	209M	native	Mistral AI
TimeH2O	108M	native	H2O.ai; live from 28 October, with covariates from 29 October
t0-alpha, t0-beta, t0-gamma	102M, 256M, 256M	native	The Forecasting Company
Ensembles: quarter-hourly median of fixed members, point and each quantile; live from 20 October			
`ens_fm`			the newest history-only model of five native-covariate families
`ens_fm_cov`		native	their five covariate variants
`ens_hybrid`		native	those five and `lgbm_prod`
References: the grid operators' forecasts			
`tso_cutoff`			as published before the gate; rated only where it covers every selected series; live only[e]
`tso_final`			final day-ahead wind and solar, published after the gate; never ranked[f]

[a] *Full*: every NWP model's columns, the ensemble statistics and differences, the calendar and, for price, yesterday's price. The foundation models take a compact subset instead: three or four ensemble means and spreads per target, and for price the load forecast, fuel cost and the renewable drivers of the coupled zones.

[b] Retrained daily for price and load, monthly for wind and solar.

[c] Every foundation model runs on the series' own history; those with a covariate entry run a second time with the frozen covariates (e.g. `chronos2` and `chronos2_cov`).

[d] Covariate variants retired after trial runs: covariates made Moirai 2 worse, and Toto gained too little for its run time and memory.

[e] Elia's 11:00 day-ahead wind and solar forecast for BE, or, where that field is still empty at the run, Elia's most recent forecast read at the run; NED's national wind, solar and load forecast for NL, fetched at the cut-off; the ENTSO-E day-ahead load forecast elsewhere.

[f] Elia 18:00, RTE 16:15, ENTSO-E for DE-LU and Danish wind, NED in the evening for NL.

CES, TBATS) were in the field in its first two weeks and were retired: on 15-minute data with a 96-step daily cycle and a 28-day context they were not competitive. Their records are kept but no longer shown.

3.6 Scoring

For every forecast, issue day d and series s the benchmark records MAE, RMSE and bias and, for models with deciles, the Winkler score and coverage of the 80% interval and the weighted interval score (WIS). For the median and the four central intervals that the nine deciles define, the WIS equals twice the pinball loss averaged over the deciles [6],

$$\mathrm{WIS} = \frac{2}{9\,T} \sum_{t=1}^{T} \sum_{\tau \in \{0.1,\ldots,0.9\}} \max\big(\tau\,(y_t - q_{\tau,t}),\ (\tau-1)(y_t - q_{\tau,t})\big), \tag{2}$$

with T the scored quarter-hours of the day. Errors are compared across series of different scale by relating them to the baseline `naive_weekly`, and they are aggregated *before* they are normalised: for model m, series s and the set $\mathcal{D}_{m,s}$ of issue days on which both m and the baseline were scored,

$$\mathrm{RelMAE}_{m,s} = \frac{\sum_{d\in\mathcal{D}_{m,s}} \sum_t |e^{m}_{d,t}|}{\sum_{d\in\mathcal{D}_{m,s}} \sum_t |e^{\text{base}}_{d,t}|}, \qquad \mathrm{RelRMSE}_{m,s} = \Big(\frac{\sum_{d,t}(e^{m}_{d,t})^2}{\sum_{d,t}(e^{\text{base}}_{d,t})^2}\Big)^{1/2}, \tag{3}$$

and Rel. WIS likewise as the ratio of the summed daily WIS of model and baseline. A ratio of means rather than a mean of daily ratios keeps one calm day, on which the baseline happened to be nearly right, from dominating the average. Over several series the per-series values are combined by the mean, the median or the geometric mean, selectable on the benchmark's website. The geometric mean, used in the tables below, treats halving and doubling the error symmetrically and penalises a model that fails on one series.

Three choices in these definitions affect which model is ranked best, and they are stated here explicitly. *The functional rewarded.* Absolute error is minimised by the median of the predictive distribution and squared error by its mean [22], and for skewed targets such as solar power or spiking prices the two differ. Most foundation models report the median of their predictive quantiles (Chronos, TiRex, Moirai, Sundial, Toto, t0, Timestral, Tabby). TimesFM, the climatology and the LightGBM and ridge entries, fitted on squared error, report a mean. The standings use MAE and therefore favour the median. Rel. RMSE, which favours the mean, is reported alongside. Bias is reported separately, since neither measure separates a systematic offset from random error. *The benchmark in the denominator.* Every scale-free error measure is defined relative to a benchmark forecast [23]. EnergyBench uses the value of the same quarter-hour one week earlier, a seasonal naive forecast that reproduces the daily and weekly cycles of price and load and requires no fitting. Unlike the scaling of the MASE, which uses the naive forecast's error on the training data, the baseline's error is taken on the same days as the model's, so that both face the same conditions [24]. For wind and solar this benchmark is weak, and relative errors well below one are to be expected. It serves as a common unit of comparison and is not a measure of how difficult a series is to forecast. *Aggregation.* Errors are averaged over the 96 quarter-hours of the delivery day and over issue days within a series, and then across series with equal weight per series. With equal weights, the standings measure performance across forecasting problems rather than where most energy is at stake.

3.7 Ranking

A *race* is one issue day and one series; its field is every model with a scored forecast in it. A model whose run failed is left out of that race rather than ranked last (the run tracker on the site shows every failure), and a model joins the races of the series it declares from the day it is added. Leaving failures out is defensible only because the benchmark runs every entry itself: an entry cannot choose the days on which it forecasts. A benchmark that accepted submitted forecasts would have to count a missing forecast as

a loss, or replace it by a penalised baseline, as Predico does [25], since a participant could otherwise withhold forecasts on difficult days. The grid operators' pre-gate forecast, which covers only some series, is rated only on selections whose every series it forecasts (for example BE wind, or load); on any other selection it takes no part in the races, so that no rating rests on a different set of series from the rest of the field, and its errors are shown without a rating. Within a race every pair of models plays one game, won by the lower relative error of the day (equal errors split the game). All games of the selected period are fitted at once with a Bradley–Terry model [34], $P(i \text{ beats } j) = p_i/(p_i + p_j)$, by the minorisation–maximisation algorithm of Hunter [35], and reported on the Elo scale, $\mathrm{Elo}_i = 1000 + 400 \log_{10} p_i$, with the geometric mean of the strengths fixed at 1 so that 1000 is the field's average. One virtual drawn game against an average opponent keeps the rating finite for a model that wins or loses every game. The construction follows rating systems for chess and sports [8]. Unlike a running Elo, the rating does not depend on the order of the races, and a win over a strong model counts more than a win over a weak one; the same construction underlies the Chatbot Arena leaderboard [36].

The 95% interval is the 2.5th–97.5th percentile of 200 refits on issue days drawn with replacement. Days rather than races are resampled because all races of a day share the same weather and tend to be won and lost together. Overlapping intervals mean the order of two models is not settled. Any pairwise rating is relative to the field: a model's score depends on whom it played, and several near-identical entries (for example three releases of one model family) take wins from each other and from everyone else. The pooled relative errors do not depend on the field and are always reported next to the rating. A model is listed as provisional if it has fewer races than the smaller of 12 and a fifth of the most any model has. The rating can be computed for any selection of zones, targets and issue days. The rating counts wins and not their margins (Section 5).

3.8 Operation and reproducibility

The benchmark runs as scheduled jobs on one server: the live run at 11:30, scoring at 07:00, 14:45 and 20:45, an hourly fetch of provisional actuals, and a catch-up at 03:30. Every forecast, input snapshot, covariate table, provenance file and revision log is kept, and the website is a static export of them, so any number on it can be recomputed. The configuration file that defines the benchmark (cut-off, gate, resolution, deciles, context, baseline, sources, admission lags) is versioned with the code: changing it changes the benchmark. The website and this paper report live results only.

4 Results: the first thirty live days

The results below are those of the live record only, scored on the forecasts as they were locked before each gate. After the coverage of the record, they give the standings, the value of the covariates for foundation models, the role of model size, and the comparison with purpose-built models and with the forecasts issued by the grid operators.

4.1 Coverage

The live record covers issue days 2–31 October 2026, the last day with every target scored when this paper was written. Series enter the record when they can be scored on a stable definition, and entries when they are added, so the record is uneven (Figure 4). The Dutch wind, solar and load record starts on 11–12 October, with NED's national series (Section 3.3), and DE-LU, DK1 and DK2 join on 15 October.[3] ENTSO-E has published Danish load only in part since 10 October, which leaves ten scored days, and the pre-gate reference covered only part of the delivery day on ten zone-days (Section 5). A series with so few days would weigh as much as any other in the averages below, so Danish load is left out. The record therefore holds 22 of the 24 series, with between 14 and 30 issue days each. Because the entries also joined at different times, every comparison below is made on the races the compared models share.

[3] Danish wind lost three issue days: 17 and 18 October to its own change of source, and 24 October, whose delivery day was that of the change to winter time, to a missing actual.

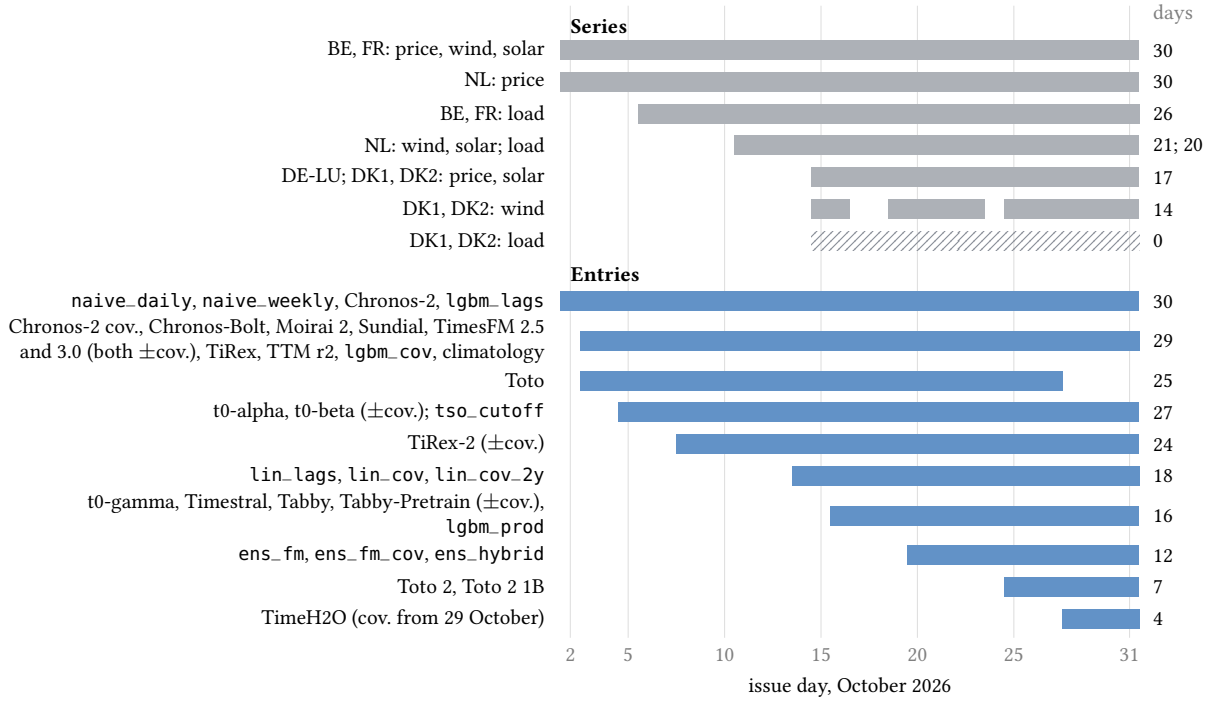


Figure 4: The live record, issue days 2–31 October 2026. Top: when each series is scored from, with its number of scored issue days (Danish wind lost three; Danish load is forecast but left out, Section 4.1). Bottom: when each entry first forecast, with the issue days it ran; Toto left after 27 October.

4.2 Standings

Table 5 gives the standings for all 42 entries, and Figure 5 shows their ratings. The two median ensembles of covariate models, live since 20 October, rate highest (`ens_hybrid`, Elo 1276 and Rel. MAE 0.233; `ens_fm_cov`, 1234 and 0.242). Among single models, t0-gamma, Timestral, TimesFM 3.0 and Chronos-2 with covariates lead (Elo 1190–1214, Rel. MAE 0.273–0.297), with overlapping intervals. Behind the leading group the field splits in two. Apart from TimesFM 2.5 and TimeH2O, whose four days give it a wide interval, every covariate-aware foundation model is separated by its interval from every history-only model. The best history-only entries, TimeH2O over its four days, Toto 2 at 1B parameters over its seven and the history-only ensemble (Elo 1042, 1003 and 981), rank behind at least fifteen covariate models. Among the history-only foundation models the differences are small (pooled Rel. MAE 0.53–0.79, leaving out Toto 2 and TimeH2O) compared with the effect of giving any of them the weather (0.27–0.37 with covariates, Table 6).

The ensembles are medians with members fixed in advance by a rule, not by score (Table 4), so they do not require knowing which member will perform best. On the twelve issue days on which they ran (20–31 October, 262 races each), `ens_fm_cov` has a lower error than each of its five members (0.242, against 0.246–0.276), and `ens_hybrid`, which adds `lgbm_prod`, a model of a different class trained on the same weather, has a lower error than every single model (0.233). Twelve days are, however, too few to rank the ensembles reliably against the best members. TimeH2O shows why the pooled errors of entries that ran on different days cannot be compared. With covariates it has the lowest pooled error in Table 5 (0.157), but over its three days, 29–31 October, which were easy for every model: on the same races the leading entries reach 0.145–0.158 (`ens_hybrid` 0.145, `lgbm_prod` 0.149, TimesFM 3.0 0.157). Its rating (1195, interval 1000–1330) places it among them, with an interval that reflects its few races (Section 5).

The grid operators' pre-gate forecast (`tso_cutoff`) has no rating in these standings. It covers only 8 of the 22 series, so a rating would rest on games in those series alone, and under the rule of Section 3.7 it takes no part in the races of a selection it does not fully cover; the ratings of the other entries are

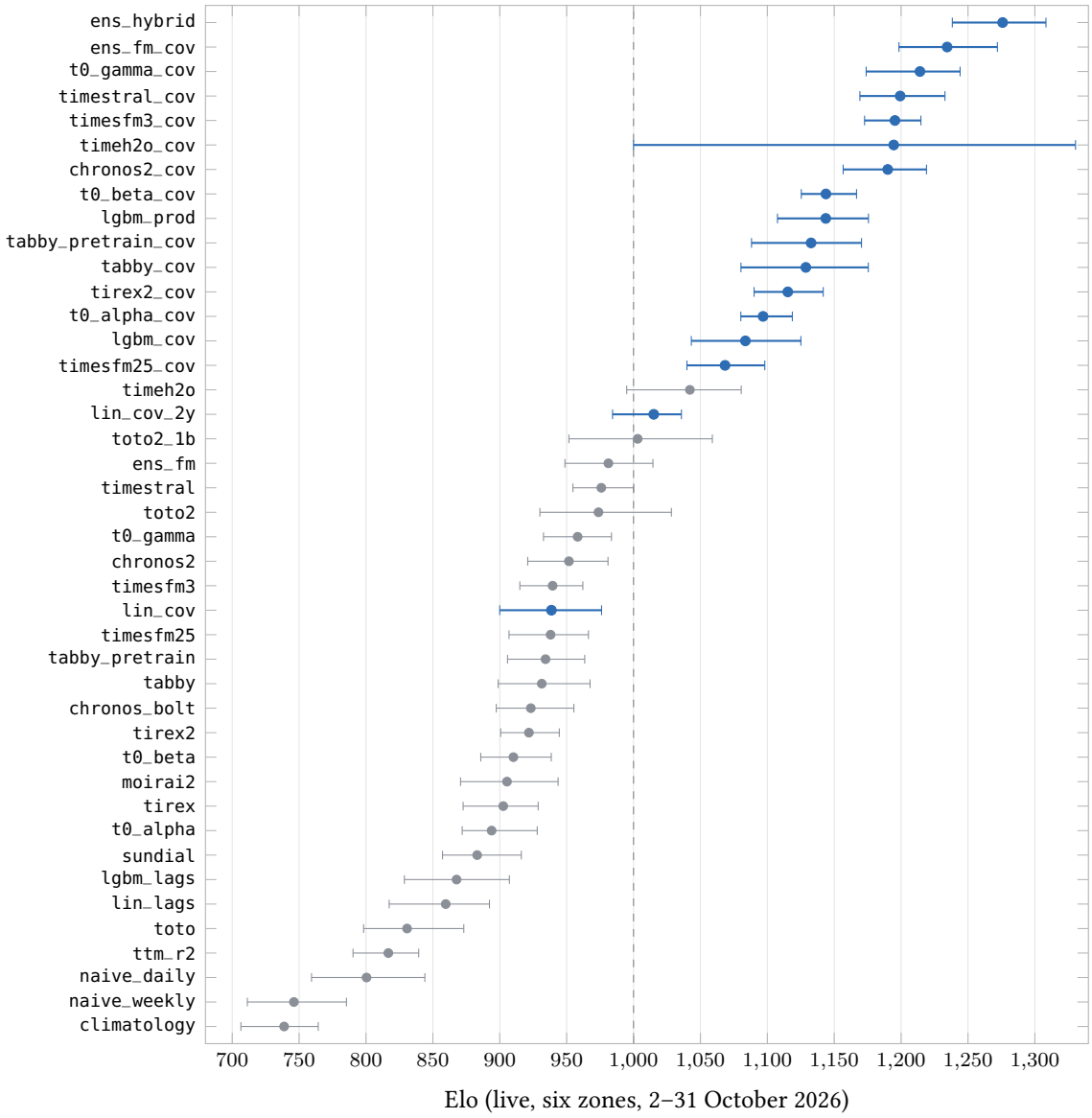


Figure 5: Live Elo ratings with day-bootstrap 95% intervals, entries of Table 5. Blue: models using the frozen covariates; grey: history only. The pre-gate TSO forecast is not rated over all series (Section 4). The dashed line is the field average.

fitted without it. It is listed below the ranked entries with its pooled errors, which are a mean over a different set of series, weighted towards load, and cannot be compared directly with those of the entries that forecast all 22. A rating would also count wins rather than margins: on the 93 races they share, `ens_hybrid` has the lower error in only 48% of them, and its pooled error is level with that of the operators. Section 4.6 therefore compares the operators with each entry on the races both forecast.

Figure 6 breaks the pooled error down by series. The covariate models lead on wind and solar in every zone, by a margin much larger than the differences between them; Dutch and Danish solar are the hardest of these series. On load the history-only foundation models are level with the covariate ones, and the grid operators' pre-gate load forecast is strong only in France.

4.3 Effect of covariates on foundation models

Ten foundation models run both on the series' own history and with the frozen covariate table, so the value of the weather can be measured model by model on exactly the same races (Table 6; TimeH2O,

Table 5: Live standings, issue days 2–31 October 2026, all six zones (22 series). Elo with its day-bootstrap 95% interval; Rel. MAE and Rel. RMSE pooled per series against `naive_weekly`, geometric mean over the series a model forecasts ("all") or over the zones of one target. All entries; covariate models in bold.

	Model	Elo	95% interval	Races	Rel. MAE	Rel. RMSE	Price	Wind	Solar	Load
					all series		Rel. MAE per target			
1	**ens_hybrid**	1276	[1238, 1308]	262	0.233	0.265	0.202	0.128	0.290	0.513
2	**ens_fm_cov**	1234	[1198, 1272]	262	0.242	0.275	0.203	0.135	0.302	0.534
3	**t0_gamma_cov**	1214	[1174, 1244]	346	0.274	0.307	0.234	0.165	0.323	0.574
4	**timestral_cov**	1199	[1169, 1233]	346	0.273	0.303	0.261	0.152	0.305	0.590
5	**timesfm3_cov**	1195	[1173, 1215]	481	0.283	0.306	0.237	0.163	0.332	0.663
6	**timeh2o_cov**†	1195	[1000, 1330]	66	0.157	0.178	0.073	0.052	0.452	0.527
7	**chronos2_cov**	1190	[1157, 1219]	481	0.297	0.328	0.238	0.186	0.353	0.642
8	**t0_beta_cov**	1144	[1125, 1167]	467	0.322	0.357	0.275	0.197	0.388	0.647
9	**lgbm_prod**	1144	[1108, 1176]	346	0.293	0.309	0.271	0.141	0.365	0.702
10	**tabby_pretrain_cov**	1133	[1088, 1170]	346	0.313	0.339	0.275	0.194	0.358	0.638
11	**tabby_cov**	1129	[1080, 1175]	346	0.314	0.339	0.272	0.196	0.367	0.630
12	**tirex2_cov**	1115	[1090, 1142]	442	0.312	0.329	0.271	0.165	0.388	0.722
13	**t0_alpha_cov**	1097	[1080, 1119]	467	0.337	0.358	0.346	0.173	0.378	0.740
14	**lgbm_cov**	1084	[1043, 1125]	481	0.339	0.353	0.317	0.156	0.376	1.035
15	**timesfm25_cov**	1068	[1040, 1098]	481	0.367	0.392	0.319	0.199	0.405	0.981
16	timeh2o†	1042	[995, 1081]	88	0.326	0.396	0.144	0.232	0.696	0.590
17	**lin_cov_2y**	1015	[984, 1036]	380	0.396	0.408	0.290	0.186	0.453	1.609
18	toto2_1b	1003	[952, 1059]	154	0.485	0.543	0.275	0.415	0.822	0.652
19	ens_fm	981	[949, 1014]	262	0.495	0.528	0.336	0.489	0.679	0.564
20	timestral	976	[955, 1000]	346	0.528	0.556	0.351	0.552	0.695	0.603
21	toto2	974	[930, 1028]	154	0.498	0.556	0.289	0.406	0.849	0.691
22	t0_gamma	958	[933, 984]	346	0.543	0.572	0.355	0.535	0.741	0.662
23	chronos2	952	[921, 981]	488	0.586	0.617	0.380	0.564	0.884	0.645
24	timesfm3	939	[915, 962]	481	0.595	0.620	0.373	0.590	0.880	0.675
25	**lin_cov**	939	[900, 976]	380	0.464	0.480	0.388	0.207	0.575	1.469
26	timesfm25	938	[907, 966]	481	0.592	0.625	0.383	0.554	0.827	0.760
27	tabby_pretrain	934	[906, 964]	346	0.563	0.591	0.382	0.589	0.731	0.638
28	tabby	931	[899, 968]	346	0.571	0.599	0.373	0.586	0.798	0.630
29	chronos_bolt	923	[897, 955]	481	0.607	0.636	0.387	0.575	0.914	0.700
30	tirex2	922	[901, 945]	442	0.586	0.606	0.388	0.535	0.866	0.694
31	t0_beta	910	[886, 938]	467	0.601	0.625	0.412	0.546	0.899	0.670
32	moirai2	905	[871, 944]	481	0.618	0.651	0.413	0.547	0.915	0.758
33	tirex	903	[873, 929]	481	0.621	0.644	0.407	0.575	0.832	0.845
34	t0_alpha	894	[872, 928]	467	0.614	0.637	0.408	0.570	0.860	0.766
35	sundial	883	[857, 916]	481	0.636	0.641	0.430	0.590	0.942	0.713
36	lgbm_lags	868	[829, 907]	488	0.687	0.709	0.497	0.580	0.927	0.919
37	lin_lags	860	[817, 892]	380	0.628	0.615	0.453	0.504	0.912	0.815
38	toto	831	[798, 873]	393	0.790	0.813	0.556	0.763	0.966	1.040
39	ttm_r2	817	[790, 839]	481	0.692	0.673	0.455	0.529	0.929	1.245
40	naive_daily	800	[759, 844]	488	0.825	0.877	0.517	0.634	0.995	1.855
41	naive_weekly	746	[711, 785]	488	1.000	1.000	1.000	1.000	1.000	1.000
42	climatology	739	[706, 764]	481	1.007	0.961	1.147	0.649	0.901	1.895
References: the grid operators' forecasts (Section 4.6)										
–	tso_cutoff	–	–	182	0.374	0.390	–	0.139	0.350	0.634
–	tso_final	–	–	–	0.175	0.193	–	0.115	0.267	–

Races differ because entries joined at different times (Toto 2, 25 October; ensembles 20 October; t0-gamma, Timestral, Tabby, `lgbm_prod` 16 October) or left (Toto, 27 October). The references are not ranked. `tso_cutoff` forecasts 8 of the 22 series (BE and NL wind and solar; load in BE, NL, FR and DE-LU), so it has no Elo over all series, and its pooled errors are not comparable with those of the ranked entries (Section 4). `tso_final`, published after the gate, is shown for wind and solar in BE, NL, FR and DE-LU; its Danish wind forecasts cover a different scope from the actuals and are left out (Section 4.6). † TimeH2O ran on the last four issue days only, with covariates on the last three, which were easy for every model; its pooled errors are not comparable with those of the other entries (Section 4.2).

whose covariates reduce its error by 34% on the 66 races of its three days with both variants, has too few to include). All ten gain from the covariates. For the seven that take covariates natively, the pooled relative error falls by 45–52%. The reduction is 64–72% on wind and 55–62% on solar, where a model that sees only the last four weeks of output has no information about tomorrow's wind or cloud cover,

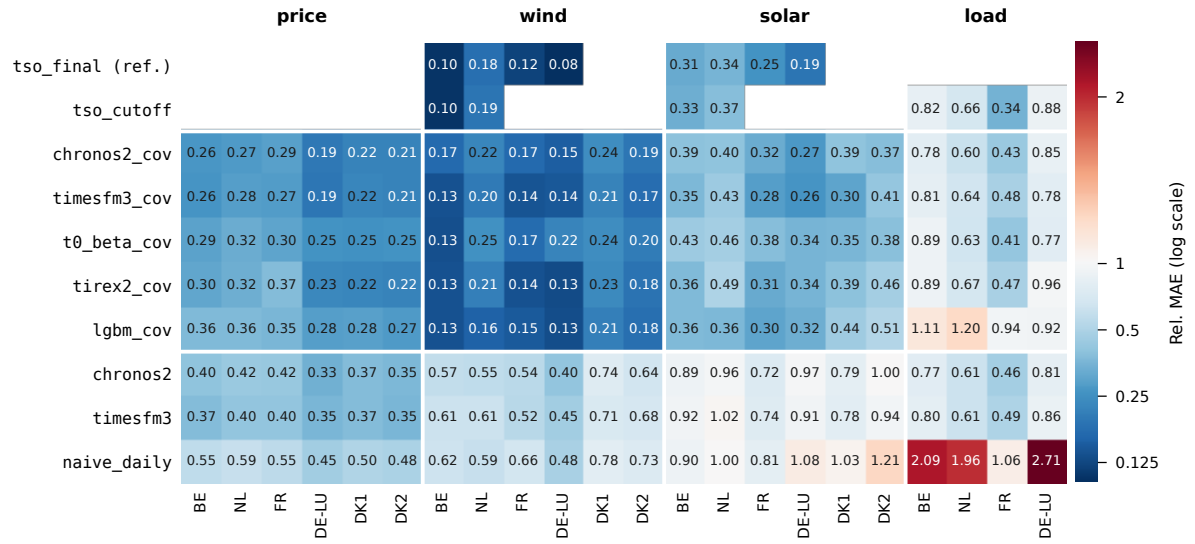


Figure 6: Pooled Rel. MAE per series for the references, five covariate models, two history-only foundation models and `naive_daily` (live, 2–31 October 2026; colour on a log scale, 1 is the seasonal-naive baseline). Empty cells: no forecast, or (for `tso_final` in DK1 and DK2) a forecast of a different scope.

Table 6: The same foundation model without and with the frozen covariates, on the races both forecast (live, all zones, issue days 2–31 October 2026). "Native": the model takes known-future covariates itself; "ridge": an in-context ridge regression on the covariates, the model forecasting its residual (Tabby forecasts load without covariates). Rel. MAE pooled per series, geometric mean over series; a change is the geometric mean of the per-series ratios minus one.

				Rel. MAE		Change with covariates				
Model	Covariates	Days	Races	without	with	all	price	wind	solar	load
Chronos-2	native	29	481	0.585	0.297	−49%	−37%	−67%	−60%	+0%
TimesFM 3.0	native	29	481	0.595	0.283	−52%	−36%	−72%	−62%	−2%
t0-alpha	native	27	467	0.614	0.337	−45%	−15%	−70%	−56%	−3%
t0-beta	native	27	467	0.601	0.322	−46%	−33%	−64%	−57%	−3%
t0-gamma	native	16	346	0.543	0.274	−50%	−34%	−69%	−56%	−13%
TiRex-2	native	24	442	0.586	0.312	−47%	−30%	−69%	−55%	+4%
Timestral	native	16	346	0.528	0.273	−48%	−26%	−72%	−56%	−2%
TimesFM 2.5	ridge	29	481	0.592	0.367	−38%	−17%	−64%	−51%	+29%
Tabby	ridge	16	346	0.571	0.314	−45%	−27%	−67%	−54%	+0%
Tabby-Pretrain	ridge	16	346	0.563	0.313	−44%	−28%	−67%	−51%	+0%

and 15–37% on price, through the load forecast, fuel cost and the expected renewable infeed. On load the change is small (−13% to +4%), since a load curve is dominated by daily and weekly cycles that the history already contains. Of the three models that receive the covariates through an in-context ridge regression, the two Tabby models gain nearly as much (44–45%). TimesFM 2.5 gains less (38%), and its load error rises by 29%, because the ridge regression fitted on 28 days extrapolates poorly.

The effect does not depend on the zone (Figure 7, where every point of a zone is computed on the same races). In every zone the covariates reduce Chronos-2's error by 50–67% on wind and solar and by 31–42% on price, while on load they change it by between −6% and +5%. On wind and solar the grid operators' forecasts are ahead of Chronos-2 with covariates in every zone where they exist, and `lgbm_cov` beats both of them in NL; on load, Chronos-2 with covariates is ahead of the pre-gate forecast in BE, NL and DE-LU and behind it in FR.

Figure 8 illustrates the difference on two delivery days on which the covariates helped more than

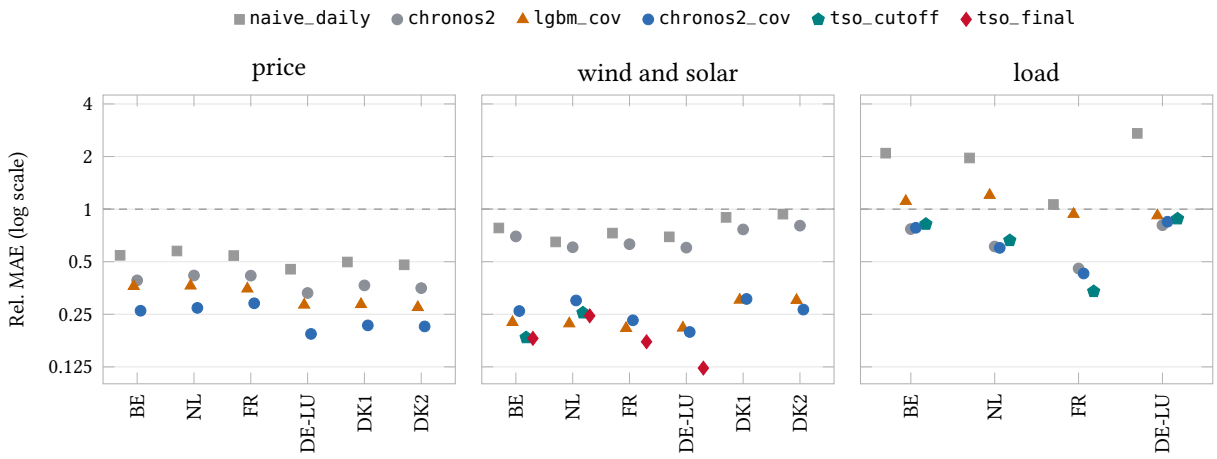


Figure 7: Pooled Rel. MAE per zone and target group (live, 2–31 October 2026, log scale; the dashed line is the seasonal-naive baseline). Every point of a zone and panel is computed on the same races: those that all models shown there forecast, the TSO references included where they exist (`tso_cutoff`: BE and NL wind and solar, and load; `tso_final`: wind and solar outside Denmark, Section 4.6). Danish load is left out (Section 4).

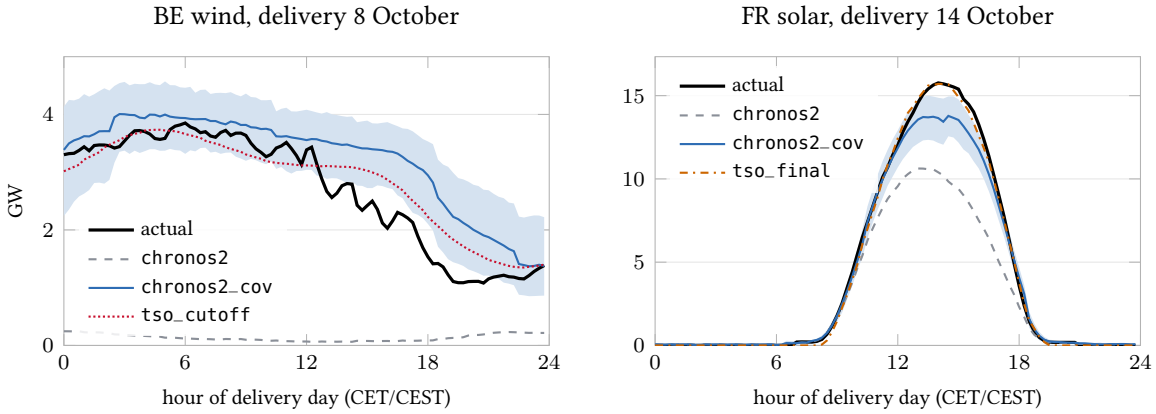


Figure 8: Two delivery days: actual output, Chronos-2 without and with covariates (shaded: the covariate version's 10–90% band), and a TSO forecast (Elia's pre-gate forecast for BE; RTE's final forecast, published after the gate, for FR).

in the median race. In the median wind or solar race, Chronos-2 with covariates has 41% of the error of Chronos-2 without; on these two days the shares are 21% and 28%. On 8 October Belgian wind generation was between 3.1 and 3.9 GW until midday and fell to about 1.1 GW by the evening. Chronos-2 on its own history carried its recent output forward and forecast less than 0.3 GW for the whole day. The covariate version followed the forecast weather, although, like Elia's pre-gate forecast, it anticipated the evening decline too late. On 14 October French solar peaked at 15.8 GW: Chronos-2 alone reached 10.6 GW, the covariate version 13.8 GW, and RTE's final forecast, published after the gate, 15.7 GW.

4.4 Model size

Figure 9 sets each foundation model's error against its number of parameters, on the sixteen issue days 16–31 October, on which fifteen of them ran (346 races each; Toto left the field on 27 October, and Toto 2 and TimeH2O joined after the 16th). Without covariates, larger models do somewhat better (Spearman $\rho = -0.52$, $p = 0.05$), but the spread is small and not monotone: Timestral (209M), t0-gamma (256M) and Chronos-2 (120M) lead the rest by at least 0.02, TimesFM 3.0, the largest at 331M, sits mid-field,

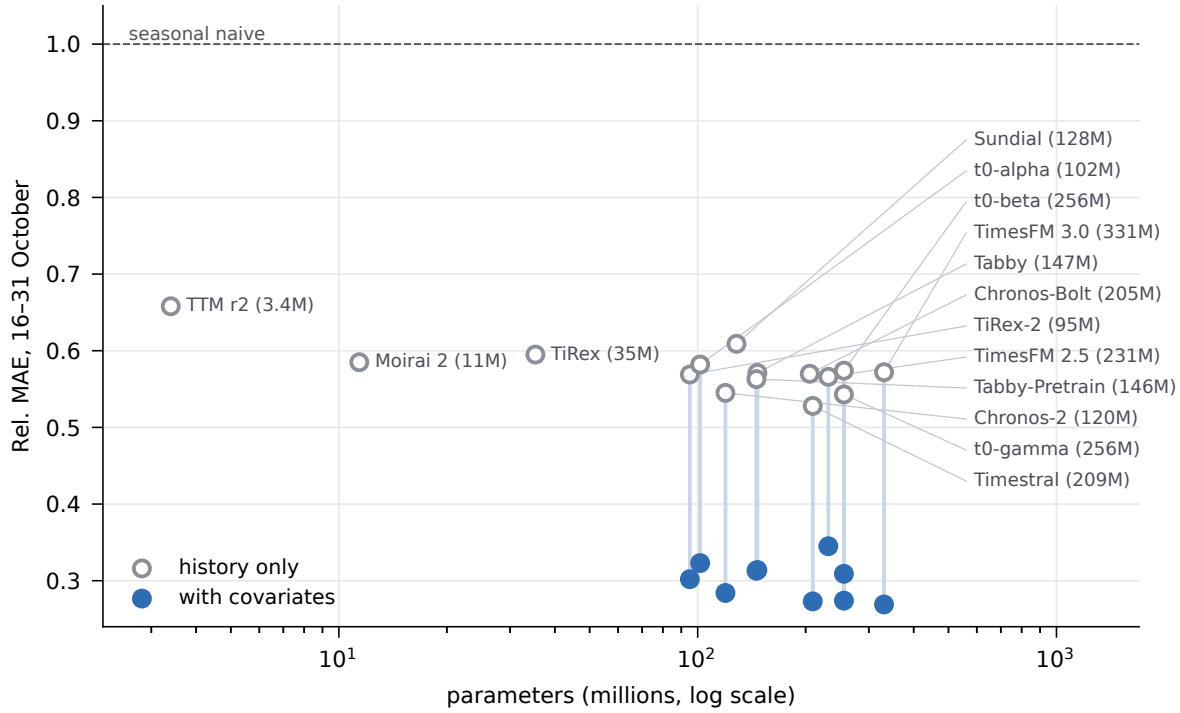


Figure 9: Pooled Rel. MAE against parameter count for fifteen foundation models, without covariates (open) and, where a variant exists, with the frozen covariates (filled, joined to the same model without), on the issue days 16–31 October 2026 (all zones, 346 races per model). Parameter counts are those of the checkpoints the benchmark loads; TiRex-2 uses about 38M of its 95M in univariate mode.

and TTM r2 at 3.4M is the weakest. Above about 100M the relation disappears ($\rho = -0.19$, 11 models), and it is not significant with covariates ($\rho = -0.38$, ten models), where TimesFM 3.0, Timestral and t0-gamma lead Chronos-2 by 0.01–0.02. Every covariate model, whatever its size, has a lower error than every history-only model, so the weather input has a larger effect than any difference in size.

4.5 Against purpose-built models

The covariate-aware foundation models are used zero-shot, yet of the five that take covariates natively and have run over almost the whole record, four have a lower error than a LightGBM model trained on the same covariate table, on the races each shares with it (`lgbm_cov`, Rel. MAE 0.331–0.339, against 0.283–0.322), and t0-alpha is level with it (0.337 against 0.338). LightGBM receives more of the table: every NWP model's columns, the calendar and, for price, yesterday's price, whereas the foundation models receive a small set of ensemble means and spreads. LightGBM is the best of them on wind (0.154–0.156, against 0.163–0.197) and level on solar (0.357–0.376, against 0.332–0.388), but considerably worse on load (1.025–1.035), where 28 days are too few to learn the response of load to the weather. Trained on two years, `lgbm_prod` does better. On the days both ran (16–31 October) it reaches 0.293, against 0.322 for `lgbm_cov`, within the range of the native-covariate foundation models (0.269–0.323). The direction agrees with FETS [14], where covariate-informed zero-shot foundation models also outperformed gradient-boosted models trained on the target data, although on fixed historical test blocks rather than live.

4.6 Against the grid operators

The grid operators' forecasts are compared with each entry on exactly the races both forecast. Two references are used. The pre-gate forecast (`tso_cutoff`) is the operators' forecast as published before

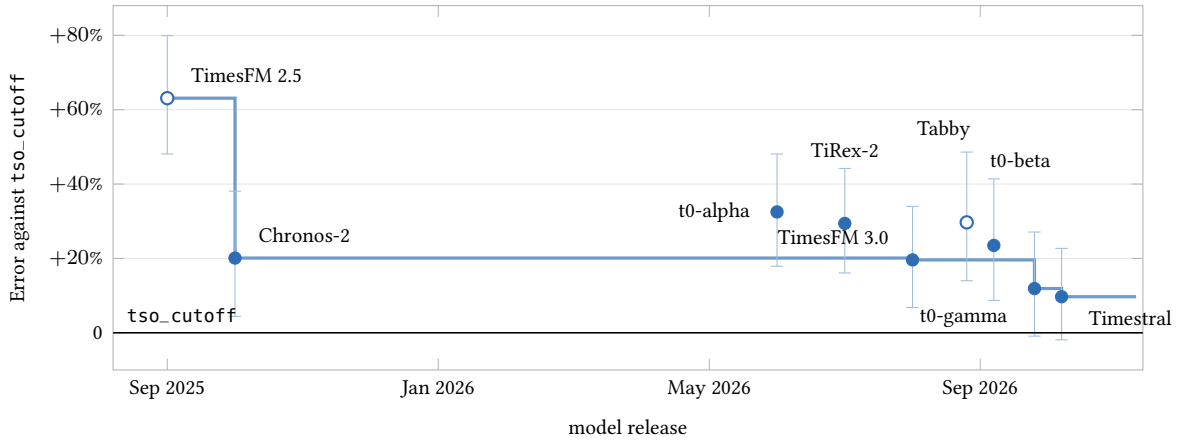


Figure 10: Foundation models with covariates against the grid operators' pre-gate forecast, by model release date, with day-bootstrap 95% intervals; the line is the best model released so far. Open markers: covariates through a ridge regression. Geometric mean of the per-series error ratios on the same 125 races (issue days 16–31 October 2026, eight series); every model ran on the same days, so the figure compares model generations, not progress over time.

the gate, and is therefore information that a market participant could have used. The final forecast (`tso_final`) is published between 16:15 and 18:00 on $D-1$, after the gate. It is based on later weather runs and more recent measurements than any entry can use, and serves as an upper reference rather than as a competitor.

Table 7 and Figure 10 set the foundation models with covariates against `tso_cutoff` on the same 125 races: the sixteen issue days 16–31 October, BE and NL wind and solar, and load in BE, NL, FR and DE-LU. The operators remain ahead overall. The closest single models, Timestral and t0-gamma, have 10% and 12% more error (95% day-bootstrap intervals -2 to $+23$ and -1 to $+27$). TimesFM 2.5, which takes the covariates through a ridge regression and gains least from them (Table 6), has 63% more. Within a model family the newer release has the lower error on these races: TimesFM 3.0 has 27% less error than TimesFM 2.5 (interval 22–31) and t0-gamma 16% less than t0-alpha (8–22).

Figure 10 places the same gaps against each model's release date, with the best model released so far as a frontier. The frontier falls from $+63\%$ (TimesFM 2.5, September 2025) to $+20\%$ a month later (Chronos-2) and to $+10\%$ (Timestral, October 2026). Only the first step is larger than the intervals; the later releases improve on Chronos-2 by amounts within them. All models ran over the same weeks, so the figure compares model generations on a common test, not the progress of one forecaster over time. TimesFM 2.5 and Tabby take the covariates through a ridge regression, which may explain part of their distance.

The gap differs by target. On load the closest models are level with the operators overall (Timestral $+0.4\%$, t0-gamma -2%): they have 7–16% less error in BE, NL and DE-LU, and 30–45% more in FR. On wind and solar they have 20–28% more error. There `lgbm_prod`, trained on two years of these series, comes closest (4% more error, interval -7 to $+14$), but it has 19% more error on load. Over the twelve days on which the ensembles ran (93 races), `ens_hybrid` is level with the operators overall ($+2\%$, interval -11 to $+18$), with 17% more error on wind and solar and 11% less on load. On wind and solar most of the improvement over a history-only model comes from the weather input. On the races of Table 7, TimesFM 3.0 reaches 0.56 of the baseline's error on wind from its history alone and 0.16 with covariates, against 0.12 for the operators; on solar the values are 0.89 and 0.39, against 0.29. On load it barely changes, from 0.66 to 0.65, against 0.59. Outside NL the pre-gate load reference is the ENTSO-E load forecast, which the load models received as an input until 15 October.

The final forecasts, published after the gate, are further ahead. On wind and solar in BE, NL, FR and DE-LU (90 races each) they reach Rel. MAE 0.115 and 0.268, against 0.138 for `lgbm_cov` on wind and 0.327 for TimesFM 3.0 on solar, the best entries on those races. Over its 16 days `lgbm_prod` comes

Table 7: Foundation models with covariates and `lgbm_prod` against the pre-gate TSO forecast `tso_cutoff`, on the same 125 races (issue days 16–31 October 2026; BE and NL wind and solar, load in BE, NL, FR and DE-LU): extra pooled error, geometric mean over series, for all eight series with its day-bootstrap 95% interval and per target group. Negative: the entry has the lower error.

Model	All	95% interval	Wind, solar	Load
Timestral	+10%	[−2, +23]	+20%	0%
t0-gamma	+12%	[−1, +27]	+28%	−2%
TimesFM 3.0	+20%	[+7, +34]	+29%	+11%
Chronos-2	+20%	[+4, +38]	+40%	+3%
t0-beta	+24%	[+9, +41]	+42%	+7%
TiRex-2	+29%	[+16, +44]	+36%	+23%
Tabby	+30%	[+14, +49]	+57%	+7%
t0-alpha	+33%	[+18, +48]	+42%	+23%
TimesFM 2.5	+63%	[+48, +80]	+62%	+65%
`lgbm_prod`	+11%	[+1, +22]	+4%	+19%

closest on wind (0.119, against 0.111). Danish wind is left out of this comparison because Energinet's final forecast and the actuals do not measure the same quantity: its day-ahead forecast for DK1 wind has run at about 0.8 of measured output for two years, a difference in level rather than in skill.

5 Limitations

The record is short and covers one season. Electricity series are not stationary: autumn has neither the solar peaks of summer nor the demand peaks of winter, and installed capacity and market rules change from year to year. Which models lead may differ between such regimes, so the standings are provisional and will change as the record grows. The three zones added on 15 October have seventeen issue days, and entries that joined late are compared with the rest only on the days they share.

The weather inputs are three deterministic models from three centres (DWD, NOAA and ECMWF), and their spread stands in for forecast uncertainty. All three are served by one provider, Open-Meteo, which is a single point of failure: on six of the thirty live days it served an ICON-EU run before the run was fully ingested.[4] The two-year training frames match the lead time of the live runs but take ICON-EU and GFS from a previous-runs product that can contain runs initialised after the cut-off of the day they describe; this concerns the training data of two entries, never what is scored.

A fair evaluation depends on details that are rarely examined: the formal definition of each actual and of each operator forecast, and the way the published data are read. The record has several examples, each documented and, where the forecast could not be repaired, voided. On seven issue days in DK1 and three in DK2, Energinet's load forecast on ENTSO-E ended at about 11:00 on the delivery day when the run read it, with a closing zero that was parsed as zero load for the rest of the day. Those `tso_cutoff` forecasts are voided, since the later hours, published after the gate, cannot repair them. Until the parser was fixed on 31 October the same zeros reached the load forecast input of the price models; those forecasts are scored as they were run (Danish price errors show no visible harm on those days, but the counterfactual was not run). The Dutch targets changed definition during the record (Section 3.3), the Danish final forecasts measure a different scope from the actuals (Section 4.6), the `tso_cutoff` reference exists only where a TSO publishes before the gate (BE and NL wind and solar, and load), and the weather points rest on public capacity registers of uneven detail (municipal data in BE, NL and DK, commune centres in FR, unit positions in DE-LU). A shared, explicit definition of each target, its source

[4] On 3, 4, 5, 10, 13 and 17 October the 06 UTC ICON-EU run was served at the cut-off with missing values over the delivery day. Models built on the three-model ensemble means were barely affected; models that read each NWP model's own columns lost accuracy on those days. A completeness check has rejected such runs since 19 October; the affected days are scored as they were run.

and its scope, recorded with the scores, makes such differences visible before they reach a ranking.

Each entry's configuration (its inputs, context length and settings) was fixed before it went live, so no entry was tuned on the days on which it is scored; the configurations were, however, chosen by authors who had seen the data of the preceding weeks. The covariate models are deliberately simple and untuned; `lgbm_cov` and `lin_cov` train on the 28-day context for comparability with the foundation models, a known handicap that `lgbm_prod` removes. A model that fails is left out of the races it missed (Section 3.7), so a model that fails often is flagged by the run tracker rather than penalised in the rating.

Pooled errors and the rating answer different questions. A pooled error, even scaled by the seasonal-naive baseline, depends on the days it is pooled over, since some days are harder than others for every model, so two entries can be compared on it only over the same days: TimeH2O with covariates has the lowest pooled error in Table 5 because its three days were easy for every model, and on those races it is level with the leaders (Section 4.2). The rating compares entries within each race, where the difficulty of the day is shared, and connects entries that ran on different days through the opponents they have in common, so it places the whole field on one scale, with wider intervals for entries with fewer races. The rating counts wins, not their margins. A model that beats another by a wide margin gains as much from the game as one that wins narrowly, and the rating does not compare the margin a model achieves with the margin expected from the two ratings. The pooled relative errors, reported next to the rating, carry the margin: `ens_hybrid` has the lower error than the operators' pre-gate forecast in only 48% of their races, while its pooled error is level with theirs (Section 4.6). Rating systems in sports, from which the race format is borrowed, have been extended to margins of victory, either by scaling each update with the margin, as in FiveThirtyEight's Elo ratings [9], or by modelling the margin itself [10, 11]; least-squares ratings of the Massey type [12] fit the margins directly. A rating of this kind on the relative errors should be considered in later versions.

The forecasts are fixed before the outcome is known, but the analyses in Section 4 were not registered in advance. Their windows follow from the dates on which entries joined, and the authors chose the comparisons with the record in view. This is the third problem of the introduction, in reduced form: the scores, forecasts and analysis scripts are public, so each comparison can be repeated on other windows, but only a set of comparisons fixed before the data arrive would remove the freedom entirely. Finally, the results describe only the entries in the field.

6 Conclusion

A live benchmark aligned with the market gate excludes look-ahead and fixes the evaluation days before the outcomes are known. EnergyBench runs such a benchmark daily for the European day-ahead market, on quarter-hours, from frozen public data with recorded provenance, with rankings reported with intervals. Over the first thirty live days in six zones, the weather information available at the cut-off mattered more than the choice of model. The same foundation model forecasts with about half the error when it is given that information (for Chronos-2, in every zone). Zero-shot foundation models with covariates match or outperform a gradient-boosting model trained on the same covariate table, and the median of several such models has a lower error than any of them. Model size has a much smaller effect. The grid operators' pre-gate forecasts remain ahead on wind and solar, where the best single foundation models have 20–28% more error, while on load the two are level; the operators' final forecasts, published after the gate, are further ahead. Further zones, references and settlement rules are planned.

References

[1] S. Makridakis, E. Spiliotis, V. Assimakopoulos. The M4 competition: 100,000 time series and 61 forecasting methods. *International Journal of Forecasting* 36(1):54–74, 2020.

[2] R. Godahewa, C. Bergmeir, G. I. Webb, R. J. Hyndman, P. Montero-Manso. Monash time series forecasting archive. *NeurIPS Datasets and Benchmarks*, 2021.

[3] T. Aksu et al. GIFT-Eval: a benchmark for general time series forecasting model evaluation. arXiv:2410.10393, 2024.

[4] M. Meyer, S. Kaltenpoth, K. Zalipski, O. Müller. Rethinking evaluation in the era of time series foundation models: (un)known information leakage challenges. arXiv:2510.13654, 2025.

[5] H. Hewamalage, K. Ackermann, C. Bergmeir. Forecast evaluation for data scientists: common pitfalls and best practices. *Data Mining and Knowledge Discovery* 37:788–832, 2023.

[6] J. Bracher, E. L. Ray, T. Gneiting, N. G. Reich. Evaluating epidemic forecasts in an interval format. *PLOS Computational Biology* 17(2):e1008618, 2021.

[7] Electricity Maps. electricitymaps-contrib: data parsers and source documentation. `https://github.com/electricitymaps/electricitymaps-contrib`, accessed October 2026.

[8] A. E. Elo. *The Rating of Chessplayers, Past and Present.* Arco, New York, 1978.

[9] N. Silver. Introducing NFL Elo ratings. FiveThirtyEight, 4 September 2014.

[10] S. Kovalchik. Extension of the Elo rating system to margin of victory. *International Journal of Forecasting* 36(4):1329–1341, 2020.

[11] L. Szczecinski. G-Elo: generalization of the Elo algorithm by modeling the discretized margin of victory. *Journal of Quantitative Analysis in Sports* 18(1):1–14, 2022.

[12] K. Massey. Statistical models applied to the rating of sports teams. Honours thesis, Bluefield College, 1997.

[13] O. Shchur et al. fev-bench: a realistic benchmark for time series forecasting. arXiv:2509.26468, 2025.

[14] M. Obermeier, M. Pruckner, F. Haselbeck, A. Zeiselmair. FETS benchmark: foundation models enable scalable and generalizable energy time series forecasting. *Energy and AI*, 100867, 2026. arXiv:2604.22328.

[15] A. Garza, R. Rosillo, R. Mendoza-Smith, D. Salinas, A. R. Williams, A. Ashok, M. Goswami, J. M. Juárez. Impermanent: a live benchmark for temporal generalization in time series forecasting. arXiv:2603.08707, 2026.

[16] T. Hong, P. Pinson, S. Fan, H. Zareipour, A. Troccoli, R. J. Hyndman. Probabilistic energy forecasting: Global Energy Forecasting Competition 2014 and beyond. *International Journal of Forecasting* 32(3):896–913, 2016.

[17] M. Meyer, S. Kaltenpoth, H. Albers, K. Zalipski, O. Müller. TS-Arena: a live forecast pre-registration platform. *KDD '26: Proceedings of the 32nd ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, 2026. arXiv:2512.20761.

[18] M. Kleinebrahm, J. Berrisch, P. Eiser, W. Fichtner, V. Hagenmeyer, M. Hertel, N. Koster, S. Lerch, R. Mikut, J. Priesmann, M. Schienle, B. Schäfer, J. Weinand, F. Ziel. Energy-Arena: a dynamic benchmark for operational energy forecasting. *22nd International Conference on the European Energy Market (EEM)*, Trondheim, 2026. arXiv:2604.24705.

[19] T. Hong, P. Pinson, Y. Wang, R. Weron, D. Yang, H. Zareipour. Energy forecasting: a review and outlook. *IEEE Open Access Journal of Power and Energy* 7:376–388, 2020.

[20] R. Weron. Electricity price forecasting: a review of the state-of-the-art with a look into the future. *International Journal of Forecasting* 30(4):1030–1081, 2014.

[21] J. Lago, G. Marcjasz, B. De Schutter, R. Weron. Forecasting day-ahead electricity prices: a review of state-of-the-art algorithms, best practices and an open-access benchmark. *Applied Energy* 293:116983, 2021.

[22] T. Gneiting. Making and evaluating point forecasts. *Journal of the American Statistical Association* 106(494):746–762, 2011.

[23] R. J. Hyndman, A. B. Koehler. Another look at measures of forecast accuracy. *International Journal of Forecasting* 22(4):679–688, 2006.

[24] A. Davydenko, R. Fildes. Measuring forecasting accuracy: the case of judgmental adjustments to SKU-level demand forecasts. *International Journal of Forecasting* 29(3):510–522, 2013.

[25] Predico collaborative forecasting platform, INESC TEC with Elia and 50Hertz. `https://predico-elia.inesctec.pt`.

[26] Commission Regulation (EU) No 543/2013 of 14 June 2013 on submission and publication of data in electricity markets.

[27] ENTSO-E Transparency Platform. `https://transparency.entsoe.eu`.

[28] Open-Meteo, single runs and previous runs APIs. `https://open-meteo.com`.

[29] M. Löning, A. Bagnall, S. Ganesh, V. Kazakov, J. Lines, F. J. Király. sktime: a unified interface for machine learning with time series. arXiv:1909.07872, 2019.

[30] A. F. Ansari et al. Chronos: learning the language of time series. *TMLR*, 2024.

[31] A. Das, W. Kong, R. Sen, Y. Zhou. A decoder-only foundation model for time-series forecasting. *ICML*, 2024.

[32] G. Woo et al. Unified training of universal time series forecasting transformers. *ICML*, 2024.

[33] A. Auer et al. TiRex: zero-shot forecasting across long and short horizons with enhanced in-context learning. arXiv:2505.23719, 2025.

[34] R. A. Bradley, M. E. Terry. Rank analysis of incomplete block designs: I. The method of paired comparisons. *Biometrika* 39(3/4):324–345, 1952.

[35] D. R. Hunter. MM algorithms for generalized Bradley–Terry models. *Annals of Statistics* 32(1):384–406, 2004.

[36] W.-L. Chiang et al. Chatbot Arena: an open platform for evaluating LLMs by human preference. *ICML*, 2024.